

Catecholamines

Dopamine
Norepinephrine
Epinephrine

Learning objectives

Neurochemistry
of
Catecholamines

- Synthesis
- Release
- Inactivation

Dopamine &
Norepinephrine
Systems

- Anatomy (cell groups & pathways)
- Receptors
- Behavior

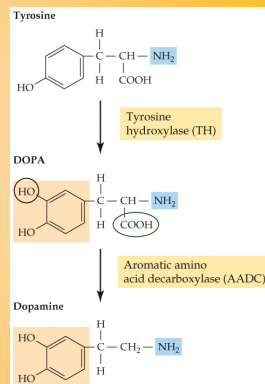
Catecholamines are synthesized from the amino acid tyrosine

Tyrosine

- is obtained from the diet

Enzymatic synthesis

- Tyrosine hydroxylase (TH)
- Fe^{2+} , O_2 , BH_4
- AADC
- Pyridoxal phosphate (B_6)



Pharmacological
tools used to
determine the
role of DA in
behavior

Drug	Mechanism of action
Dihydroxyphenylalanine (-DOPA)	Dopamine (DA) precursor
Carbidopa	Peripheral aromatic amino acid decarboxylase (AADC) inhibitor
Phenelzine and tranylcypromine	Nonselective monoamine oxidase (MAO) inhibitors
Moclobemide	Selective MAO-A inhibitor
Selegiline and rasagiline	Selective MAO-B inhibitor
Entacapone, opicapone, and tolcapone	Catechol-O-methyltransferase (COMT) inhibitors

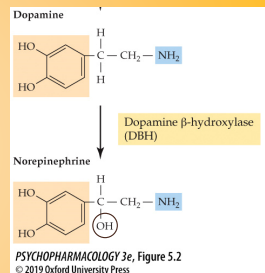
Catecholamines are synthesized from the amino acid tyrosine

Enzymatic synthesis

- Dopamine β -hydroxylase (D β H)
- Cu²⁺, O₂, Ascorbic acid

Epinephrine

- is converted from NE by PNMT
- S-adenosylmethionine (SAM)



TH is a critical control point in catecholamine synthesis

TH is the rate-limiting enzyme

TH is regulated by end-product inhibition

- More DA, less TH activity

TH is activated by phosphorylation

- Increased neural activity increases TH activity
- PKA, PKC/ERK, CaMKII
- Ser 40, Ser 31 & Ser 19 in the regulatory domain of TH

Catecholamines are loaded into vesicles by VMAT

Vesicular monoamine transporter (VMAT)

- Different from DAT located on nerve terminal
- Blocked by reserpine
- Energy provided by a proton gradient
 - ATP-driven H⁺ pump acidifies vesicle lumen

Pharmacological tools used to determine the role of DA in behavior

Drug	Mechanism of action
α -Methyl-para-tyrosine (AMPT)	Tyrosine hydroxylase inhibitor
Reserpine	Irreversible VMAT inhibitor
Tetrabenazine, deuterabenazine, and valbenazine	Reversible VMAT inhibitors
6-Hydroxydopamine (6-OHDA) and MPTP	DA neurotoxins
Amphetamine*	Catecholamine releaser and reuptake blocker
Cocaine*	Catecholamine reuptake blocker

Release of catecholamines is decreased by autoreceptor activation

Dopamine D2 autoreceptors

- Inhibition of voltage-gated Ca^{2+} channels
- Activation of voltage-gated K^{+} channels

Noradrenergic α_2 autoreceptors

- α_2 agonist clonidine; α_2 antagonist yohimbine
- clinically important in opioid withdrawal and anxiety / stress-related disorders

Inactivation mechanisms are important targets of psychoactive substances

Reuptake

- DA transporter (DAT)
- NE transporter (NET)
- Na^{+} /MA-ATPase

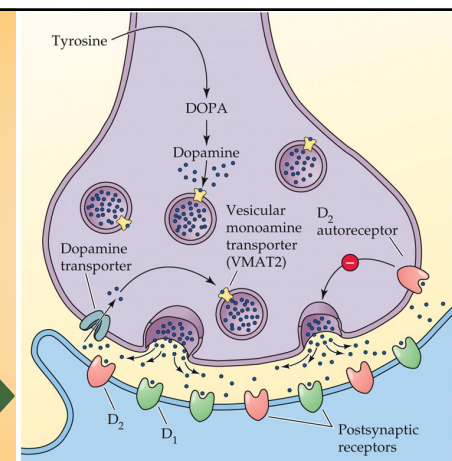
Enzymatic degradation

- Monoamine oxidase (MAO)
- Catechol-O-methyl transferase (COMT)

Pharmacological tools used to determine the role of DA in behavior

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Summary:
Synthesis,
Release &
Inactivation



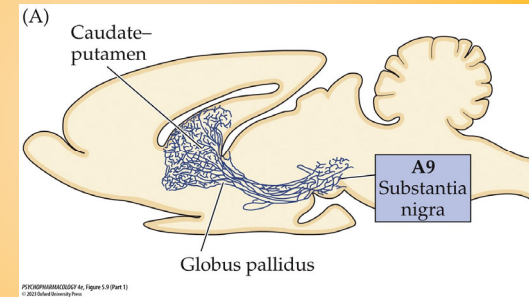
PSYCHOPHARMACOLOGY 4e, Figure 5.5
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The Dopamine System Organization and Function

There are 2 primary groups of DA neurons, that give rise to 3 pathways

Substantia nigra (A9)

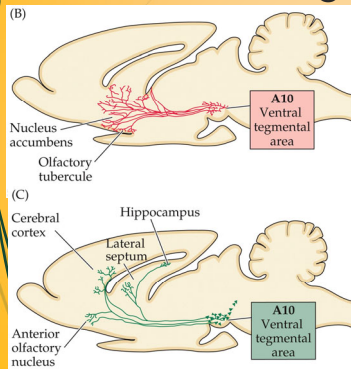
- Projects to the striatum
- Nigrostriatal pathway



There are 2 primary groups of DA neurons, that give rise to 3 pathways

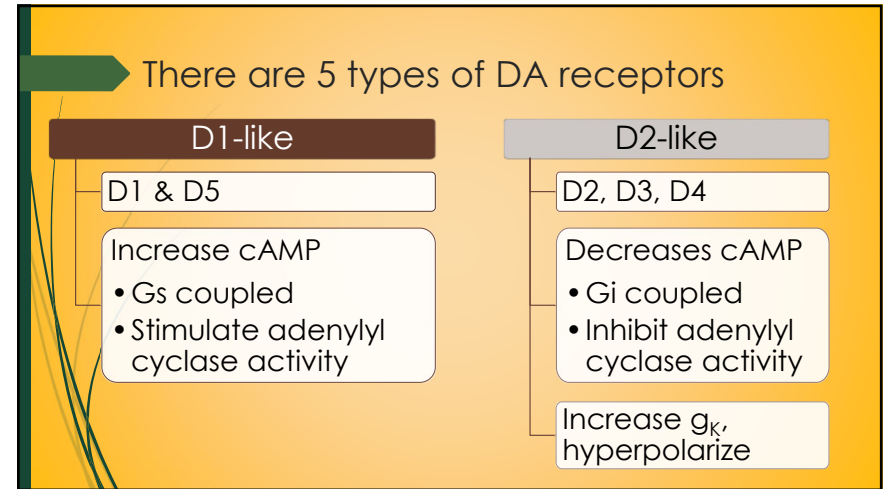
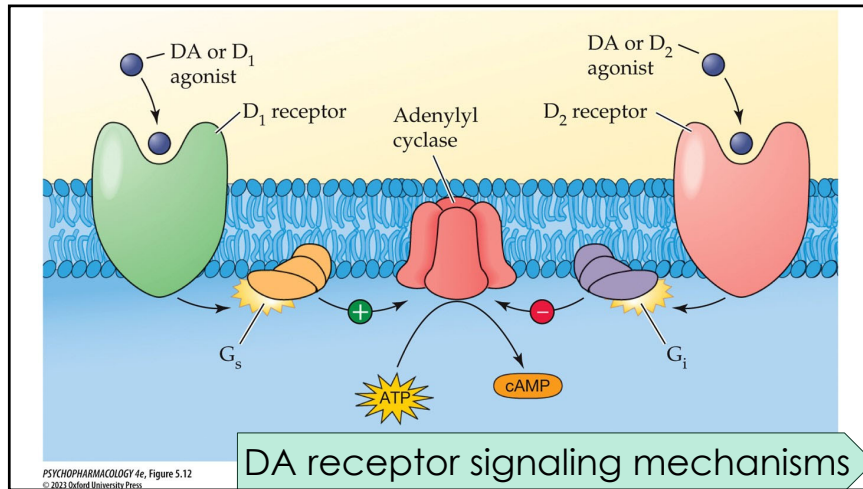
Ventral tegmental area, VTA (A10)

- Projects to limbic structures and cortex
- Mesolimbic pathway
- Mesocortical pathway



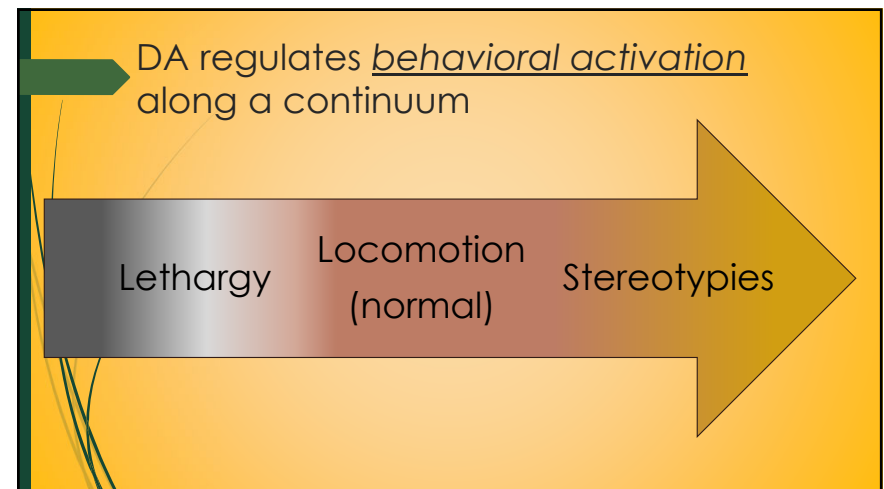
DA cell groups and pathways

Substantia Nigra (A9)	VTA (A10)	VTA (A10)	Hypothalamus (A12)
Nigro-striatal	Meso-limbic	Meso-cortical	Tubero-hypophyseal
• Activation • Motivation • Cognition	• Behavioral arousal • Reward learning	• Attention • Working memory	• Prolactin release



Pharmacological tools used to determine the role of DA in behavior

Drug	Mechanism of action
Methylphenidate*	Catecholamine reuptake blocker
Apomorphine	Nonselective D ₁ and D ₂ receptor agonist
SKF 38393	D ₁ receptor agonist
DETQ	D ₁ receptor positive allosteric modulator
Quinpirole	D ₂ /D ₃ receptor agonist
Bromocriptine, pramipexole, ropinirole, and rotigotine	D ₂ /D ₃ receptor agonists
SCH 23390	D ₁ receptor antagonist
Haloperidol	D ₂ receptor antagonist



Reducing DA synthesis or release impairs behavioral activation

AMPT blocks TH activity

- Associated with relapse into depression following positive response to antidepressant medication

Reserpine blocks VMAT

- Sedation in animals
- Depression in humans



Neurotoxic damage to DA neurons impairs behavioral activation

6-OH-DA selectively damages DA neurons

Bilateral central lesions induce:

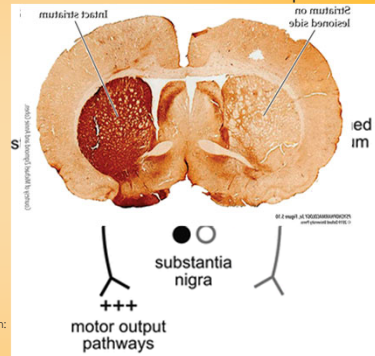
- Sensory neglect
- Motivational deficits (cease eating or drinking)
- Parkinsonian motor deficits

Unilateral 6-OHDA lesions to the SN_{pc}

Ipsilateral rotation

Loss of DA input to lesion side

DA releases behavior on the *intact* side



Dunnett S.B., Torres E.M. (2011). Rotation in the 6-OHDA-Lesioned Rat. In: The E. Dunnett S. (Eds.), Animal Models of Movement Disorders. Chromethods, 61, Humana Press

Behavioral deficits in KO mice are generally similar

TH^{-/-} knock-out (KO)

- Lethal – Loss of NE results in abnormal heart development

Dopamine-deficient (DD) mice

- Deficient feeding, drinking, grooming, locomotor behaviors
- Deficits can be rescued by restoring DA to the nigro-striatal pathway

DA cell groups and pathways

Substantia Nigra (A9)	VTA (A10)	VTA (A10)	Hypothalamus (A12)
Nigro-striatal	Meso- limbic	Meso- cortical	Tubero- hypophyseal
• Activation • Motivation • Cognition	• Behavioral arousal • Reward learning	• Attention • Working memory	• Prolactin release

D1 and D2 receptor agonists promote behavioral activation

Apomorphine

- D1 / D2 agonist
- Hyper-locomotion
- Behavioral **stereotypies**
 - Intense sniffing
 - Head bobbing
 - Licking, biting

SKF 38393

- Selective D1 agonist
- Promotes grooming

Quinpirole

- D2 / D3 agonist
- Increases exploratory behaviors – locomotion, sniffing

DA receptor antagonists suppress spontaneous behaviors (catalepsy)

Haloperidol

- D2 antagonist
- Cataleptic
- Antipsychotic

SCH 23390

- D1 antagonist
- Cataleptic

Loss of DA transmission results in behavioral super-sensitivity due to receptor up-regulation

6-OHDA lesions

- Ipsilateral rotation
- Contralateral rotation when challenged with apomorphine

Chronic D2 antagonist

- Haloperidol

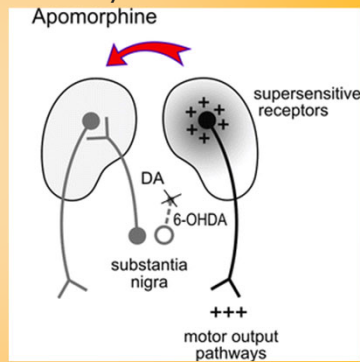
Loss of DA transmission results in behavioral super-sensitivity

Contralateral rotation

Behavioral super-sensitivity

D2 receptor up-regulation

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DA regulates behavioral activation

sedation, catalepsy

AMPT

reserpine

6-OH-DA

SCH23390, D1
haloperidol, D2

hyperlocomotion,
behavioral stereotypes

amphetamine

cocaine,
methylphenidate

Apomorphine, D1/D2

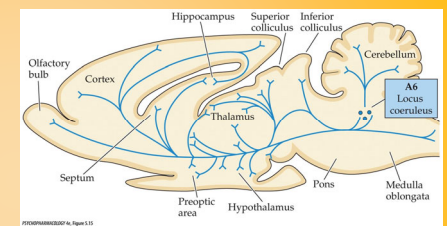
SKF38393, D1
quinpirole, D2/D3

The Noradrenergic System Organization and Function

Central NE arises from the locus
coeruleus (LC)

Locus coeruleus
(A6)

- nearly all the NE innervation to
 - cortex
 - diencephalon
 - limbic system
 - cerebellum
 - spinal cord



Sympathetic NS

- autonomic ganglia
- chromaffin cells of the adrenal medulla

There are 2 groups of adrenergic receptors (AR), alpha & beta

Alpha (α)-AR

α_1 Gq-coupled to phospholipase C (PLC)

- increase $[Ca^{2+}]_i$

α_2 Gi-coupled to AC

- decrease cAMP
- terminal autoreceptor

Beta (β)-AR

β_1 -AR & β_2 -AR

Both β_1 & β_2 are Gs-coupled to AC

- increase cAMP

Pharmacological tools used to determine the role of NE in behavior

Drug	Mechanism of action
Reboxetine	Selective NE reuptake blocker
Phenylephrine	α_1 -Receptor agonist
Midodrine and metaraminol	α_1 -Receptor agonists
Clonidine and lofexidine	α_2 -Receptor agonists
Guanfacine	α_2 -Receptor agonist
Dexmedetomidine	α_2 -Receptor agonist
Isoproterenol	Nonselective β -receptor agonist
Albuterol and levalbuterol	Selective β_2 -receptor agonists
Prazosin	α_1 -Receptor antagonist
Yohimbine	α_2 -Receptor antagonist
Propranolol	Nonselective β -receptor antagonist
Metoprolol	Selective β_1 -receptor antagonist

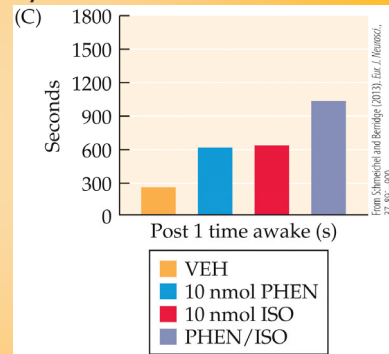
Adrenergic agonists in the LH promote wakefulness (arousal)

Phenylephrine

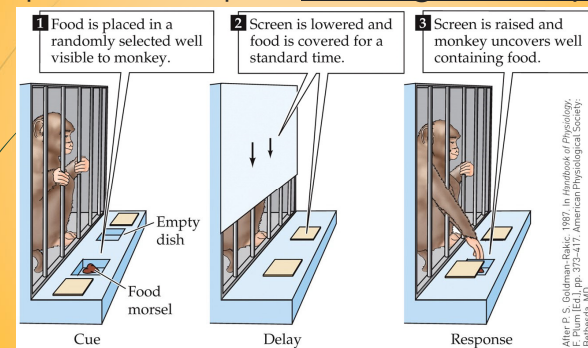
- α_1 -AR agonist

Isoproterenol

- non-selective β -AR agonist



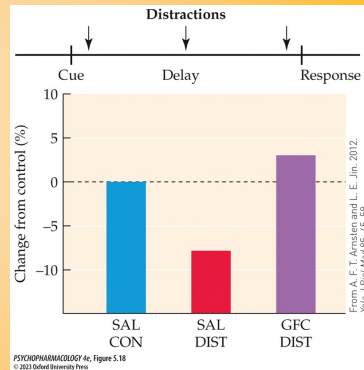
Adrenergic input to the PFC can improve or impair working memory



Adrenergic input to the PFC can improve or impair working memory

α_2 -AR agonists

- guanfacine & clonidine
- improve performance on delayed response task



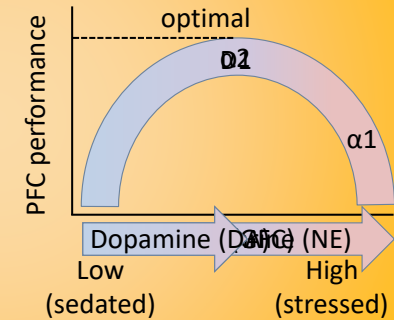
Adrenergic input to the PFC can improve or impair working memory

α_2 -AR

- High affinity for NE
- Improves performance

α_1 -AR

- Low affinity for NE
- Impairs performance



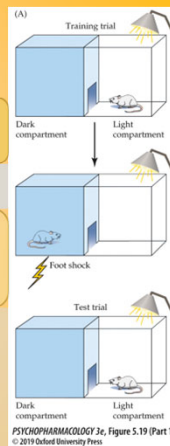
Noradrenergic control of emotional memory

Passive avoidance learning

- Foot shock induces fear / stress

EPI and GC's released from the adrenal glands; NE from LC

- EPI activates β -AR's on vagal afferents & in liver
- CNS NE potentiates central fear/anxiety mechanisms



Summary: Behavioral effects of DA-ergic & NE-ergic neurotransmission

DA

- Behavioral activation
- Goal-oriented behavior
- Working memory

NE

- Cortical arousal (e.g. wakefulness)
- Working memory
- Emotional memory